Astera Labs – Senior Executive, Network Switching Group at Marvell Technology Inc

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Specialist Background

- Over 20 years' experience in the technology industry
- Comprehensive knowledge of PCIe (peripheral component interconnect express) switches, key players, retimer content opportunities and market shares
- Well-placed to speak in detail to Astera labs' new fabric switches, competitive positioning, Nvidia GPUs (graphics processing unit) penetration and PCIe switch revenue potential

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Transcription begins

Analyst:
Astera Labs have got four product families now, like retimers, active (ph) cables, PCIe [peripheral component interconnect express] switches and CXL [Compute Express Link] controllers. Are there other specific products that you think, in terms of what you focus on, that we should centre the conversation around?

Specialist:
These are the four main product lines. Let me remind you that these product lines was something which they have conceived over a period of time. Definitely, forward-looking, they have to look at their market TAM and SAM and how do they expand and give value to their shareholders. It's a pretty young, aggressive company, and they have a good vision to solve the industry problems, which sometimes goes under the radar of the big corporate companies who look at ROI and returns and are not so risk adverse because of the commitments what a bigger corporation has got.

I wouldn't be surprised if they are looking something at scale-up switches, which is pretty prominent these days, a very hot potato topic in the industry, and nobody has really put their hands on this hot potato right now, to be honest, because if you look from an industry point of view, PCIe as such is reaching its end of life. I know people might not like this statement of mine, but it is a reality and the fact, and there are things which go with it. We are already in PCIe gen 6. PCIe gen 7, I do believe, will be one of the final entrants of PCIe, which is running at 128 Gbits per second in terms of the speed and the feed. After that, it will be more of a convergence of Ethernet. The question comes, do you need PCIe or should we just jump on the Ethernet waggon?

One of the primary drivers for this statement is also Intel. The whole driver for PCIe in the industry was Intel being the mothership institution, which was driving PCIe right from the go. This has been driven for the past, I would say, 20-plus years. As the mothership is getting into, I would say, a vintage phase and other CPUs are coming up, hyperscalers are making their own ARM CPUs plus the invent of Nvidia coming into the whole compute market segment and changing the market dynamics, the question of PCIe really becomes a big question mark in future. I'm sure this thing is playing in Astera Labs' mind as well. They have a good runway. They are still in PCIe gen 6, then the gen 7 will come. I would say they have a good runway for the next, I would say, 3-4 years. Beyond that, if they want to build themselves as a big company like Marvell or Avago or Broadcom, then they have to think beyond the spectrum.

Analyst:
PCIe gen 6 is kicking in now, gen 7 maybe to end of 2025, probably more likely 2026. That will carry you to 2027. Is 2027 the timeframe by which you think (talking over each other)?

Specialist:
Yes, '27-'28 is the timeframe where we'll also think, now what's next? I need something next. For that, if you need something in '27-'28, you need to start sowing the seeds for that kind of solution today because if you plan it today, you will start executing in '25, you will tape out in '26, you will get something in end of '26, beginning of '27. You will be prepared. You need to be prepared before the

market is prepared. That's the secret of the game. You can't be prepared after the market has made a decision about it. You have to be on the driver's seat and get to that phase.

Analyst:
Right now, they're only looking to scale out switches. My sense is PCIe scale out switches is dominated by Broadcom. It seems like it's maybe USD 1bn or so in terms of market opportunity per year, stepped up nicely with AI. How are you thinking about the market growth from here? It seems like there's going to be mixed node in terms of what you need from the AI, head node connections because SSDs [solid-state drive] are going to stay at PCIe 5. Does the gen 6 cycle here make USD 1bn become bigger? Is there a lot of growth in the scale up market opportunity from a switch perspective or if the AI step function higher has really played out and that content opportunity (talking over each other)?

Specialist:
That's a good question. I'll address one by one, and I will logically explain you the reason behind it so that you take the lay of the landscape. If there are any questions, just stop me and I'll be more than happy to zoom into it. The good news is that the market is fired up to grow. It's fired up to grow by a huge factor, and I anticipate this factor to be anywhere between 3-5x of the market size today. It's just exponentially growing. The switch market itself, if you look at it, the PCIe switch market is anywhere between USD 1bn-1.5bn, but it will grow. I will come to the reason for it to grow. Why the scale-up switching is needed is apart from the efficiency part of it is also for the return of investment. The reality of the matter is as we fine-tune these LLM models and the different workloads which are coming within the data centre, we don't know the characteristics of what utilisation will be needed for these workloads.

The main priority over here is if you don't know what the utilisation will be, but you definitely know how do you optimise this utilisation, you will be investing a certain amount of dollars in your infrastructure and you want to make sure that the dollars what you're applying in your data centre is getting utilised. The last thing you want to hear from your operations is that you invested in hardware and it was under-utilised. That is completely a loss, and we are talking about millions of dollars over here. In order to bulletproof this, in order to make sure that you have an insurance policy, that it's getting utilised, you need an inbound fabric switch. The fabric switch, what it does, in a scale-up, it bundles a couple of hardware resources. It might be the SSDs, it might be the memories, it might be the GPU, it might be the CPU. It interconnects it in, in such a way that none of them are under-utilised. If somebody is needing a more of memory, a more of GPU horsepower, a more of CPU horsepower, within the scale-up, within the pod, you are able to borrow resources.

Also, the next league is within intra-pods, you can borrow resources. We are not talking about scale-out right now, we are just talking about scale-up. Every single node of your scale-up is being fully utilised and is distributed. That is the secret of the game that at no point, you should be under resourced and starving for resources to get your job done. People don't know what category of job do they need. They don't know whether they need more of GPU, they don't know whether they need more of memory or they need more of storage. That is very dynamic because the workload, the data structures are evolving, and they really can't predict that. They can just have a metrics, they just have a performance benchmark and then they start building this system. The real utilisation is still to be fine-tuned in the future, and that will take some time in the industry to know.

What Astera and the people are doing, if you look just now focusing on PCIe switching, what they have right now, there's only one player in the market to be dominant. That is Broadcom. Broadcom had acquired this technology very, very long time back by a company called PLX Technology. PLX was the main pioneer, which worked on the PCIe switching. It's a niche market, but it is becoming more and more pervasive as we go in the AI phase. PCIe switching was not such a big, huge market. It was only used predominantly in the HPC market segment, in high-performance compute. All the national labs, supercomputing centres used to use PCIe switching. Other normal data centres and cloud vendors would not use it.

Now AI is a super set of high-performance computing, if you look at it that way. It's catching up. Since there's only one dominant player, there is a good and healthy market for a second player to come and take some of the market share. Plus, Broadcom charges a lot of its premium to its customers. It's not that it's one of the most friendly customer companies today. Definitely, Astera will get the TLC from a lot of people in the industry. That addresses how this PCIe switch market is all about.

Analyst:
I've heard PCIe market pre-AI was probably more like USD 400m. You had a step-up to USD 1bn-1.5bn, so a step-up of USD 600m-900m. Is the vast majority of that related to Nvidia HGX systems or would you say that there's already been adoption by some of the hyperscalers in terms of what they're doing in terms of internal workloads? AWS [Amazon Web Services], as an example, is someone who's been looking at PCIe scale up switches We had the AI watershed moment and the market doubled. Was that all Nvidia or were there some hyperscaler custom builds as well?

Specialist:
No, there is a hyperscaler custom build as well. You're forgetting one market segment is also the Google TPUs. There is a PCIe sitting over there also, which Google deploys in a lot of its infrastructure and data centre. That is also running on Ethernet and PCIe. A huge portion of the Google intelligence is on the TPUs, which is running in parallel with the GPUs of Nvidia and they're on PCIe. By the way, the Nvidia GPUs do not use so much of PCIe when it comes to their own platform. They are basically running on NVLink. That's been their proprietary interface since, I would say, 2010, and it is giving them the returns because there is limitation in PCIe. It do support PCIe. You can take a GPU of Nvidia and put it into x86 AMD or an Intel server. When it comes to their performance and when they build the vertical integration of DGX, they basically use the NVLink protocol. Within the brace, they have the NVLink port and they basically connect their GPUs to the NVLink port. They do not use PCIe or CXL. They do have their own tricks to get the maximum performance on their system.

The growth of this PCIe from USD 400m and seeing about USD 1bn is predominantly coming mainly from the GPUs being attached to the non-Nvidia class of clusters, that is the x86 and the AMDs. The second one where you see a PCIe coming up is basically in the flash and the storage, which is coming in. You also have accelerators in the market. The accelerators I'm talking about is like Gaudi, the Habana Labs, which Intel populated, accelerators, which are coming in the sense of TPUs, which has been designed and also a lot of custom accelerators are built by these hyperscalers. They have their custom silicon team, and these silicon teams basically put in accelerators. The fourth one, which we are missing right now is the NIC cards. All the Ethernet NIC cards, which are sitting on the server platforms today are all PCIe-based.

As the server compute power is increasing, you have basically these NIC cards which are going from 25 to 50 to 100 to 400 Gbits per second in terms of their speeds. The PCIe by itself, the value increases because now you're going from a x4 to a x8 to x16. As you keep increasing the number of lanes on PCIe, the value of it increases because that's the port density, and you pay a higher premium for those port density. Also, your cost per port of PCIe has increased and hence, the PCIe SAM and TAM has increased. One part of it is the higher deployment. The second part of it is the higher speeds and feeds, which adds more value to it.

Analyst:
There's going to be some GB200 systems that are sold that don't use the Grace CPU [central processing unit], they use x86. That will be the opportunity for some of these hyperscalers to use PCIe switches at

the end of the day. There's this 2x2 CPU to GPU [graphics processing unit] box and 36 of them are spun up together. There's 72 total. What's the switch opportunity in that?

Specialist:
There is a little bit of a twist to the story. When you're talking about 72 GPUs and these GPUs are connected to the CPUs, just look at it in this landscape and then you'll get your answer. Do you want the switch to happen at the data side of those things where the GPU is getting data and throwing data or do you want the switch to happen where the GPU and the CPU is sitting in the interface stuff? The reality of the matter is actually the switching, which you need is happening at the data level where the GPUs are talking to each other. This is not happening at the GPU to CPU level switching. You are not multitasking between the data, between the GPU and the CPU. You are not throwing the CPUs with multiple GPUs.

In reality, the switching is happening at the data level of the CPUs. If you just go to Nvidia website and a lot of their blogs, they have something called as the NVLink Switch which is basically the scale-up switch which is needed in the industry or the scale-up fabric switch. This is an Ethernet switch where all the GPUs are basically connected in mesh and the data switching to these GPUs is being handled by this fabric switch, which is called the scale-up switch. AMD started an initiative on this class of switch about four months back, which was called as a UAL, Ultra Accelerated Link.

That Ultra Accelerated Link is a direct competitor of NVLink switch, which Nvidia has it. You cannot use the NVLink switch on an Ethernet-based platform because it is completely Nvidia proprietary. That's the way Nvidia creates the pods. That's the way Nvidia does all the switching of the data within the GPUs associated with the pod, keeping the GPU to CPU link constant and very firm. It doesn't play how the CPU is handling its workload. For them, with all due respect, CPU is cheap, GPU is expensive. If CPU is cheap and they don't care about if it's being under-utilised, as long as it's not over-utilised, they are fine with it. They maintain the link between the CPU and the GPU as it is on the PCIe or NVLink, and they do the switching of all the GPUs through the Ethernet platform using an NVLink switch.

The advantage of NVLink switch, which today PCIe does not have it, is the NVLink switch does coherency between all the GPUs. It does all the aspects of scaling and pooling associated with the GPUs, where if one specific GPU cache is being under-utilised, it can borrow its cache and give it to an adjacent GPU, which is starving for cache to do some kind of a task which is assigned to it. That is the beauty of NVLink. That is why Nvidia is what Nvidia is today. This borrowing of cache can happen across 256 nodes. We are talking about 72. I'm telling you that the protocol, the NVLink can scale all the way up to 256.

Analyst:
What Broadcom is selling in terms of the PCIe switch, my understanding was if you had a box, there's the opportunity to sell 144-lane switches into a box, which has one CPU and two GPUs. Is that the right way to think about within the box, what the PCIe opportunity is?

Specialist:
Yes. You are right. Within the box, we can have two GPUs, we can have the switch and where the switching is happening, where the traffic has been split from the CPUs to these two GPUs, you can do that, yes. That's a valid case. Where basically you do disaggregation of data across two GPUs, which are on a very 2U server class of devices where there is a lot of horsepower on the CPU, and it basically is doing packet spring across two GPUs based on the workload balancing, what is happening.

Analyst:
When I think about Astera Labs coming into the market, am I right to think that it's more going to be non-Nvidia boxes? It's more like a Trainium or an Inferentia box for Amazon or whatever.

Specialist:
Most of it is going to be a non-Nvidia box. That's where it plays. They have got very close ties up with Amazon and Intel, especially Astera Labs. It can be also entertaining a hyperscaler to meet their business requirements, yes.

Analyst:
We have USD 1bn-1.5bn market, how much of that is Nvidia boxes, like 80%?

Specialist:
I think Nvidia will be anywhere between 60-70%. They're still not at the 80% benchmark. A lot of companies are still a very strong proponent of being on the x86 side. They have deep roots. Plus Nvidia is also, I wouldn't say struggling, but ramping up to meet to their scale and demand. Today, if I look at the industry, Nvidia would be in the 60-70% range.

Analyst:
You have maybe USD 300m-400m of the market that's non-Nvidia but Broadcom is not just going to fall to the wayside and let Astera take all of that. What could they likely do in terms of (talking over each other)?

Specialist:
Actually, Astera has a better leverage than Broadcom because of multiple reasons. (1) There is no price dominance over here, so they can basically leverage that. (2) They are already there in the retimers. They are already sitting in those plug points where PCIe is already sitting. In gen 6, you do need a retimer. They can basically convince the current customers that, "I can give you a bundle with my retimer as well as a switch." In that case, there is no issue of compliance. It will seamlessly work with each other because it's coming from the same institution. The third advantage what they have, Astera is completely focused on PCIe. It's like bread and butter. It's their forte. They are doing it. For Broadcom, they have hundreds of other businesses. The distractions are high. Plus price dominance is high. You know a vendor like Astera, which will be continuing to invest on the PCIe side of things, inclusive of CXL, they have a better leverage than Broadcom in terms of being in the space and establishing themselves as a market leader.

Analyst:
The only other player that I hear a lot about, it's more like a CXL hybrid is Xconn. Do they matter yet? I know that they've continued to have middling demonstrations at a bunch of these technical conferences. I imagine they'll figure out eventually. Is Xconn worth focusing on yet or are they still struggling from a development standpoint?

Specialist:
I think they're still struggling. I wouldn't put much of my attention to Xconn. The success for any company is you need a godfather company who's blessing you. Astera Labs right from the go had this blessing from Intel and Amazon. Intel and Amazon were the two companies which seed-funded Astera Labs. They were the two companies who told them, "Build this thing for me and I will make sure that you are deployed in all of my systems." At whatever state, financial state Intel is today, Intel is still Intel. It has that market presence, and it has got a lot of servers in the industry today, and it will continue to. It's not that Intel is going to go away in the next 10 years. I don't believe that. I believe somewhere something our fed, our government, Intel is a US brand, a US image. In short, it is our pride also in the compute industry. We cannot let it go away like that. It will always exist. For their existence, servers will be there. PCIe, at least I told you till gen 7 is there.

Astera has executed as a good partner to Intel and is delivering the goods what they have promised. They will keep gaining their share of the market. Broadcom is not sitting with Intel. They are not having any collaboration with Intel. Broadcom is being a 100-pound gorilla where it smells money, it goes, builds a part and starts dominating it. In this case, there is a huge technology advantage, there is a huge partner advantage, which Astera has already established.

Analyst:
For the H100 and B200 boxes that are not Grace Hopper CPU, they're x86, how do you think about Astera Lab's potential to penetrate the PCIe switch there? I feel that the Amazon opportunity is the one that's a no-brainer but what about the Nvidia boxes?

Specialist:
If these H100 and the B200 are sitting on an x86-based platform, Astera has an advantage and has their deployments there. Let me also give you a point, Astera is a very methodical organisation. It's a lean organisation. It will not build something just to create a market value for itself. It will build something where some customer has given them the direction and a promise that, "You build this. I do see a market for this, and I will use it." I would not be surprised that Amazon is building platforms with the... This PCIe fabric is optimising the utilisation of the CPUs. Having that is very much needed.

Analyst:
When you think about 2026 revenue potential for Astera Lab's PCIe switch, where are you thinking?

Specialist:
In the first year itself, they would easily be in the north of USD 50m-60m in terms of their business revenues on this product line. At 2026, they'll be in full production, yes. I would say they would be in the north of USD 70m in the revenue on this product line.

Analyst:
That's like somewhat contrary. USD 70m out of USD 1bn is 7% market share.

Specialist:
Yes. Remember one thing, this USD 70m which is coming from a USD 1bn market share, which I've told you before, it all depends upon what the port values and what the stuff is happening. I can confidently say they will cross USD 70m in their business in the PCIe side in the first ramp of their stuff. After that, it all depends upon what roadmap they are showing, what promises they are showing on the PCIe gen 7. Gen 6, the train has already been in the launch phase. There are platforms on PCIe gen 6 happening today. If you launch a product in the first year, USD 70m, this product is not expensive to manufacture by the way, the cost of this product should be somewhere in the USD 100 cost, not the selling price, the cost of this product. It should be very close to USD 100. They are making good margins on this product line for sure.

Analyst:
You're talking about the fact that people are already building these platforms but Broadcom hasn't even released silicon for its gen 6 switch chip. Should I be thinking that, "Yes, we're at gen 6 for GPUs with Blackwell but the PCIe switches themselves are largely going to be gen 5 next year?"

Specialist:
PCIe switches will be transitioning into gen 6 by 2026 timeframe. In terms of the market expectations, Astera Labs is on a very, very sweet spot.

Analyst:
I would have thought way bigger than USD 70m for two years from now if they're pretty well-positioned

for both (talking over each other).

Specialist:
I would not be overoptimistic about stuff because remember one thing that the NVLink switches are dominating over this stuff. Most of the GPUs are dominating by Nvidia. I'll tell you the other part which I see from the industry and what I have observed, this H100 and the B200s, the way Nvidia is operating, it gives the first priority for DGX. Most of their stock goes to DGX first and then it goes to the x86 world. Jensen wants to sell as much as he can, but he's more profitable when it goes under DGX because it becomes a full vertical integration. Unless and until you are somebody like Elon Musk or Meta, where he can provide it to you, but in general, he is holding back on enabling the x86 platforms, not for any reason, but he's looking at the big picture and selling it to DGX. I don't know how many of these will go to x86 and what is his long-term strategy.

Analyst:
Those boxes still have PCIe switches inside the boxes?

Specialist:
Not necessarily. DGX doesn't have PCIe switch inside it. It has a PCIe retimer, but it has a NVLink switch on it. It definitely has an NVLink switch. He's not using PCIe switch inside today. He's not taking the PCIe route at all. He doesn't want to do PCIe, so he takes the CXL route.

Analyst:
So it is USD 1bn market today fall next year if he is continuing to not prioritise PCIe switches?

Specialist:
No, but you're forgetting on the other hand, AMD is also investing a lot on their GPUs. AMD will need these PCIe switches. How will AMD solve this problem? AMD is trying to solve this problem by using a UAL switch. The UAL switch will not be in the market till 2027. The concept is just conceived this year. If somebody starts building it today, the switch will come out sometime end of next year, then it goes for field trial. There is no x86-based PCIe switch or CXL switch or UAL switch. Astera has been in the sweet spot. They'll say, "Guys, you don't have it, you don't have a choice. You have to use the PCIe switch." Broadcom doesn't have a gen 6. It all depends upon how good AMD does. If AMD systems are doing well on the GPU side, then definitely Astera Labs is on the upper hand.

AMD is collaborating today with Astera Labs. They are very much there in their launches and in their new product lines and established retimer vendor. I'm sure AMD would have gone and told them that, "You do these PCIe switch." UAL switch is a little bit more on the Ethernet side. Maybe tomorrow, Astera Labs will have some skills to do Ethernet-based switching, but today, they don't. Eventually, I would anticipate Astera Labs to come up with a UAL class of switch. PCIe will be still there for them for two years or 2.5 years.

Analyst:
Who are the top three customers for the inbox PCIe switch? It sounds like Nvidia doesn't use it, they use the NVLink.

Specialist:
The main customers, if you ask me, it will be the likes of Amazon, the likes of Intel, the likes of AMD and any hyperscaler which is heavily into the x86-based platform who are investing heavily into x86. They will be using this PCIe class of switch. Now if you look on the hyperscalers who are heavily into x86, I can say straight off the back as one is the Azure which is Microsoft. They are not into ARM servers. They do everything on the x86. The second one, which is on the front-end networks is a little bit of Meta. Meta is

a big Nvidia house, but it is on the Nvidia house for the AI and the back-end network. On the front-end network, they will be using x86. Amazon is ARM and the ones in APAC and China are also mainly going on to the ARM waggon.

Analyst:
What would you say is Astera's penetration in Nvidia GPUs? I think that at H100, they were there.

Specialist:
Heavy penetration. They endorse each other. For the retimers, Nvidia uses Astera Labs. They have a very strong collaboration. If I'm not mistaken, their GM, Alexis, is actually sitting on the board of Astera Labs.

Analyst:
I'm hearing from a lot of investors that have gone over to Asia that basically what Nvidia has done with the GB200 is, it has brought the NIC [network interface card] onto the board, the CPU tray. You don't need a retimer anymore because the distance of the data doesn't travel far enough. The CPU, GPU to NIC opportunity has gone. The CPU to GPU isn't a one-to-one management anymore. It's going through a switch and that switch, the NVSwitch has retimer capability. The retimer opportunity is really just to do with storage tray moving forward. Is it true?

Specialist:
That is true. I'll tell you why, because as you go from PCIe gen 5 to gen 6, the retimer opportunity reduces. In gen 5, you use the NRZ signals, so you need a retimer by construction. In the PCIe gen 6, you use the PAM4 signalling The PAM4 signalling is quite stronger. You don't need a retimer. I would say, by nature of the protocol, going from gen 5 to gen 6, the utilisation or use of retimer diminishes. It's not got to do whether Nvidia wanted to take it out or something. It's just the nature of the protocol that you really don't need. You will not see a much of PCIe gen 6 retimers than a gen 5 retimer's requirement. Also on the other hand, the reason why you need a retimer, and this is a little bit of a trade-off on the system level, if you use cheaper board materials and you compromise on your construction of the system, then the signal gets distorted or it gets attenuated. At that point, you need a retimer.

Either you save on the system manufacturing or you pay for the retimer. That trade-off you have to do, and that's a system house call, whether you do a calculation of buying a retimer and you do a calculation of building the PCB, which is of a good material and you get a clean signal. You're right, your analysis is absolutely right that in gen 6, with the CPUs and the NIC sitting inside, you eliminate the need of the retimer. That's an industry known stuff that in PCIe gen 6, retimers can be completely be avoided.

Analyst:
It's still needed for the storage component because the storage (talking over each other).

Specialist:
I'll tell you why it's needed for the storage because storage is a very price-sensitive market. People use cheaper IOs, cheaper drives, and they don't have a very long shelf life. They keep it for maybe three years, four years of storage, and they don't feel that there is a need for it. Since they use a very low-cost stuff to manufacture, you're not getting a premium. The dollar per gigabit every year is reducing, so they use a retimer. The second is it's an assurance that it gives you a better signalling so you know at least when you are transferring data and all, there is no loss or corruption of the data happening.

Analyst:
For a GB200 box, how many storage trays are there? How should I think about what the retimer needs would be? There are 36 boxes but I don't think every single one of them needs an individual storage

tray, they probably share it a little bit. Should I think about the entire opportunity?

Specialist:
I'll give you the best case and the worst case. If there are 36 boxes, each boxes can basically have four storage devices. 4x4 is 16 lanes. Each storage link is a four-lane link. The maximum capacity you can have is each box can carry four storage devices, built inside on an average. That's like they would basically do it. Minimum, based on their whole storage rack and their data structures, what they need, they do a trade-off between the capacity and the number of devices they hold because the more number of devices you hold, you're not only loading the system, but it's also a thermal and a heat dissipation associated with the fans running as well. You have to look at the whole system calculation. It all depends. I can't give a finite answer for this, but I can tell you the metrics that basically average four per box which will be there because that is 16, which is 4x4, each device will have a four lane of PCIe going into it, and then you can do the math from there.

Analyst:
A best case would be 4x36 and worst case would be 36 or maybe a little bit lower than that?

Specialist:
Yes, correct.

Analyst:
For Astera, that's still great because when you think about the content per server opportunity, it's increasing for them because in the past, they did maybe 8-15 and now they can do 16 or more.

For example, Trainium deployment, everything we just spoke about is not the case for them. They do CPU to GPU on a one-to-one management. You need retimers for that. Let's say that it's probably like four GPU box, so that's four retimers. They might have four retimers for the storage and 1-2 NICs. We're now at 10-ish. I heard that in the back-end for something such as Trainium, you also have GPU to GPU connections that also require retimers to some extent. Should I be thinking about 35, 40 retimers in a total Trainium box at the end of the day or is it smaller than that? How do you think about that content opportunity?

Specialist:
It depends upon the two classes of network. Either you are doing your CPU load sharing or you're doing a data load sharing. If it's a data load sharing between the GPUs to GPUs, then it basically goes on the Ethernet side of stuff and it basically goes on the back of the GPUs. That is basically the load balancing happening on the data side. In those class of networks, you will not require a retimer. It will just be doing load balancing on an Ethernet-based switch. At that point, there is no retimer needed. If it is load sharing on the CPU side, where the CPU is doing load sharing with the GPUs and doing data load spraying on the GPUs which is quite feasible. In that case, every of these GPUs connected to a CPU will require a retimer or a switch associated with it. You will have a centralised switch, which will be like an Astera switch, which is connected to all the GPUs. Then you have another side, you have multiple CPUs which are connected to the switch. The switch does the load balancing from the host to these GPUs.

The way it operates is you have this CPU device which has been leased to an organisation or has been leased to do a particular complex task. There are multiple users who are associated with this task. When these multiple users start firing their task to these CPUs, then this CPU does the load balancing that, "Okay, I'll give these tasks to multiple GPUs and then reassemble everything and give it back to the user as an end result. In that case, you will require these retimers or the switches which are associated with the GPUs. I would anticipate each of these GPUs not having more than 6-8 classes of retimers or a switch connection going to a centralised switch device or a centralised retimer device, not more than 6-8, I would say. The reason I'm saying 6-8 is because these classes of CPUs has a specific index called as spec-end, which is a performance number per user. That spec-end today is in the range of about 25-30 per user. Anything beyond that is not feasible. It is basically, then you have to reduce the number of users. This is basically a class of core, which is in the class of 32-48 core device.

Analyst:
Are you talking about it all from the fron-end of the system?

Specialist:
Yes. It doesn't matter whether it's front end or back end. If it's at the back end, still the user is basically giving this (audio distorts) back end side. Predominantly, it is in the back end because the GPUs are connected to the back end of the network.

Analyst:
From that, I understand Trainium probably is 6-8.

Specialist:
Yes.

Analyst:
What do you make of Astera telling people that they can do up to 40 in hyperscalers that are using it, both using retimers, running at the back-end?

Specialist:
Yes, they can do 40 because if you look at the switch point of view, I think so it's a 64-lane switch. The limitation is not on the Astera side. The limitation is on the CPU side that if you have so many connections, will the CPU be able to do the load balancing? I use this number of spec-end of 25-30. If you have more users, if you have 40 connections, your CPU performance will go down. That means each link will get a spec-end of about 15 or 20. It drastically gets reduced. Every of these hyperscalers has an SLA with the end user that they guarantee a certain level of performance for their CPUs. You cannot be sitting on an Amazon instance, E2 instance and see your workload being stalled and taking time for you to process your information. You will not want to go with them as your cloud vendor. Astera has done a wonderful job by scaling till 40 because they anticipate that tomorrow, the CPUs are going to get much more complex and much more powerful. If you have a product today, it should last for the next two or three years. That's why they have scaled it all the way to larger than each node can handle today.

Analyst:
What is AWS doing with Trainium and increasingly in Inferentia from an internal workload standpoint? I'd imagine they care less about spec performance because they don't have SLAs in place. Do you see a larger retimer content opportunity there where they'll take a hit on performance?

Specialist:
Absolutely, yes. If it's not directly leading with the SLA with the customers and it's the internal inference engine, then it basically all boils down to one thing, it's basically their convergence time that if they basically put to this inference, a lot of loads which are going in parallel, do they want to wait for 10 minutes or do they want to wait for five minutes? It all boils down to that. It's also the convergence time what the inference machines are taking. If they are willing to compromise on the time, then they can put in more of these switches. Basically, they have to balance between resource and time. if they optimise their resources to use these 20-30 instances on the PCIe switch, and they don't care whether it takes five minutes more, but they are happy of not investing into the infrastructure and putting more hardware resources, fine. It's just a convergence time, what is the delta. If it's not SLA, it's not customer bound,

it's internal to them, they are willing to spend that five minutes more. It's an internal operational call what they have to make.

Analyst:
Would you guess that Trainium and Inferentia, of the instances they're handling, it's probably 80-90% internal at this stage?

Specialist:
Yes, I do believe that number. Yes.

Analyst:
We talked about the CPU load share inherent limitation, probably like 6-8 retimers for external base instances. For the internal ones where you remove the SLA limitation, what does that do for the retimer content for Trainium and Inferentia?

Specialist:
The retimer content, if they can handle 40, it can be between 25-30 easily for instance. It all depends upon how they are doing their internal connectivity. Yes, it can go to 20-25 easily.

Analyst:
You never think like a management team's liar. It seems like what they're talking about from like that up to 40 opportunity is related to internal workloads for some of the hyperscaler customers, which the good thing for them is that that's the majority of the instances that they handle right now.

Specialist:
Yes.

Analyst:
Retimer market share, like Astera claims that coming into the IPO, they had north of 90% market share. I think there's plenty of debates around the fact that plenty of companies coming into the market, their market share is going to normalise at a lower level. It sounds like Broadcom is actually floundering on that front. Just because they released retimer, they thought they were going to get traction. I haven't heard a lot of data points that that's been the case. What do you think can be normalised retimer market share for Astera Labs? It's obviously not going to be 90%.

Specialist:
You have a good point. In fact, a lot of companies are entering into this space. I would say they do have a dominance and a first-mover advantage, and not only a first-mover advantage, but established credible player in this market, and things don't change so oftenly. The whole PCIe gen 5 market, they have it almost, I would say. 90% is the right number. Even in PCIe gen 6, they have a dominance because it becomes a legacy. You have already worked with the vendor, your software, your protocol, your SDK is all tuned to what the vendor has done. In PCIe gen 6, there are multiple bigger players which are coming into the market. I would say it will get normalised, but the 90% is not going to become 50% or 60%. They will still be a dominant player in the industry. I do still anticipate them to be above 70% of the market, to be honest. It's not going to vanish. Yes, there will be some market dilution happening with other players coming into it, but not to a level that Astera will go below 50%. I don't see that.

Analyst:
Given that we have bigger, more established players coming to the market, they talked about the ability to get ASP uplift at each next generation of PCIe. Do you think retimer pricing is something that's difficult to see those types of uplifts? Would you expect that actually pricing is probably pretty

flat or maybe actually down?

Specialist:
They will see an uplift because every infrastructure companies are increasing their prices because those days are gone where the prices should remain the same. Especially in the area where they are dominant, why shouldn't they increase their price? TSMC is asking us more price. The cost of labour has gone up, the cost of fabrication from TSMC and the substrate cost has gone up. To keep and maintain the margins and show more profitability, they will increase their ASPs. It's bound to happen. If they are dominant and they have already locked their customers and given them the service, they will slowly start tripling their prices.

Analyst:
For the retimers, for the GB200, we talked about that it's only going to be storage and those are x4 lanes. I think for the retimer, selling x16 lane form factors. If there's only x4 lanes on the storage side of things, is it likely that all of the storage retimer content opportunity, those will just be x8s, there's no reason to buy x16s?

Specialist:
No, you can split the x16 into 2x8s or 4x4s, by the way. You can do that. You can buy a x16 retimer and segment it into 4x4 or 2x8. You can do that, yes. It's not necessarily a x16 will be a complete x16. You can do bifurcation of the retimers.

Analyst:
If it's up to 16 lanes of storage but it might be x4 lane devices, do you only need one retimer for that? Are the storage devices far away enough where you need a retimer at each device point?

Specialist:
No. For the storage, it's recommended to put the retimer for the data integrity. You're risking your system on the data. It again depends upon the system and the platform and how much of marginality you have because, let me remind you, the storage devices get very hot. When the PVT condition changes, the data integrity has to be balanced. Retimer is just a safety lock that if your devices are getting heated, your signal integrity is being maintained at that point, at that PVT. You provide more marginality. Now coming to your question of a x4 or a x16, it is a system requirement. If there are four devices of x4, you'd rather buy one x16 device because it will be more economical, and more economical not only from a price point of view, but also from the board real estate, where you are placing these devices. You don't have to spread out and put more components on your board. You will reduce your board real estate, and it will be cheaper than buying 4x4 retimers than buying one x16 retimer.

Analyst:
My math would just be how many lanes of storage do I think they're going to be and then divide that by a x16 retimer and that's how many retimers you need.

Specialist:
Yes. Remember, you're using a factor of four lanes per storage.

Analyst:
If you have four x4 storage devices, how many retimers?

Specialist:
Let me clarify with you. Each storage device is basically a four-lane device. It has four electrical lanes, which has to be connected to the CPU. That's the x4 coming from. Now if you have four of these storage

units, you will need a total of 16 lanes. Are we correct on that math?

Analyst:
Yes, I get that.

Specialist:
Coming to your question, you can look at it as two different ways, half full or half empty. You can say, "I'm just going to buy retimer, which is going to be a x16 and connect four of these devices." In that way, if I answer your question, I'll say, you need only one retimer device and you'll have four storage devices. I can also answer and say, "No, you can have four retimer devices where each retimer device is a x4 device." It all depends on whether you are buying a retimer which is a four-lane retimer or you're buying a retimer which is a 16-lane retimer, which can handle four devices.

Analyst:
More likely that they buy the x16 because it's better priced.

Specialist:
One thing is it's better priced than buying four individual ones, that's number one. Now the other question is, which you told me or which I agreed, that these retimers are sitting on the storage devices, it's not sitting on the CPU card. If it's sitting on the storage device itself, then you can't put a x16 because each storage device will have only one device to it. You are forced to buy four of these.

Transcription ends

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